

Chapter 9 – Gmail Architecture and Mail Flow

Learning Objectives

After completing this chapter, you should be able to:

- Understand how email travels across the Internet.
 - Explain the roles of SMTP, DNS, MX, SPF, DKIM, and DMARC.
 - Understand inbound and outbound mail flow.
 - Explain Gmail routing.
 - Troubleshoot common email delivery issues.
 - Design secure enterprise email systems.
-

1. Business Problem

Imagine a company:

Contoso Ltd.

Domain:

contoso.com

Employees use Gmail for business email.

One day:

- Employees can send emails.
- Employees cannot receive emails from external domains.

Internal emails work normally.

Question:

Where would you investigate first?

Answer:

MX records

Because inbound email relies on DNS MX records to locate the correct mail server.

This is exactly the scenario we discussed during our training.

2. What is Email?

Email is a communication system where:

- The sender composes a message.
 - The sender's mail server transfers it.
 - DNS locates the recipient's mail server.
 - The recipient's mail server delivers it to the mailbox.
-

Basic Mail Flow

Sender

↓

Sender Mail Server

↓

DNS Lookup

↓

Recipient Mail Server

↓

Recipient Mailbox

Every email follows this general path.

3. SMTP

SMTP stands for **Simple Mail Transfer Protocol**.

SMTP is responsible for **sending and relaying email**.

Think of SMTP as the postal delivery truck.

SMTP is **not** responsible for reading email.

Example

Rahul sends an email.

rahul@company.com

↓

SMTP sends the email

↓

Google Mail Servers

↓

Internet

↓

Recipient Mail Server

SMTP Responsibilities

- Send email
 - Relay email
 - Retry temporary failures
 - Communicate with other mail servers
-

4. DNS

DNS (Domain Name System) translates domain names into information required by computers.

For email,

DNS tells the Internet:

"Which mail server accepts email for this domain?"

DNS contains multiple record types.

Common examples:

Record	Purpose
A	Maps hostname to IP address
CNAME	Alias to another hostname
TXT	Verification and security records
MX	Mail delivery
SPF (stored in TXT)	Sender authorization
DKIM (stored in TXT)	Email authentication
DMARC (stored in TXT)	Email policy

5. MX Records

MX stands for **Mail Exchange**.

MX records tell other mail servers:

"Deliver email for this domain to these mail servers."

Without MX records,

external servers do not know where to deliver email.

Example

```
contoso.com
```

```
↓
```

```
DNS Lookup
```



Interview Scenario

Employees can:

✓ Send email

✗ Receive external email

Internal email works.

Most likely cause?

Incorrect or missing MX records.

Why?

Outgoing mail uses Google's SMTP servers.

Incoming mail depends on external servers finding your MX records.

This matches the answer you gave during our practice.

6. SPF

SPF stands for **Sender Policy Framework**.

SPF tells receiving mail servers:

"Which servers are allowed to send email for my domain?"

SPF helps prevent sender address spoofing.

Example

Company uses:

- Google Workspace
- Mailchimp

Correct SPF should authorize both.

Example (conceptual):

SPF

↓

Google Mail Servers

+

Mailchimp Servers

If Mailchimp is missing,

emails sent through Mailchimp may fail SPF checks.

This is exactly the scenario we discussed.

7. DKIM

DKIM stands for **DomainKeys Identified Mail**.

DKIM uses **public-key cryptography** to verify that an email has not been modified during transit.

Process:

1. Sending server signs the email using a private key.
2. Receiving server retrieves the public key from DNS.
3. The signature is verified.

If the email is altered,

the signature no longer matches.

DKIM validation fails.

DKIM Flow

Google Mail Server

↓

Create Email

↓

Sign with Private Key

↓

Internet

↓

Recipient

↓

Retrieve Public Key from DNS

↓

Verify Signature

↓

Valid?

↓

Yes → Accept

No → DKIM Fail

Interview Scenario

Question:

An email is modified after leaving the sender.

What happens?

Answer:

The digital signature no longer matches.

The recipient verifies the signature using the public key.

DKIM validation fails.

This is the exact reasoning you used during practice.

8. DMARC

DMARC stands for **Domain-based Message Authentication, Reporting, and Conformance**.

DMARC tells receiving mail servers what to do if SPF or DKIM validation fails.

Policies include:

Policy	Action
none	Monitor only
quarantine	Send to spam/quarantine
reject	Reject the message

DMARC Flow

Incoming Email

↓

SPF Check

↓

DKIM Check

↓

DMARC Policy

↓

Deliver

OR

Spam

OR

Reject

Interview Scenario

Question:

The domain has:

DMARC = reject

SPF fails.

DKIM fails.

Result?

Answer:

The receiving server rejects the email.

Exactly the answer you gave during training.

9. Inbound vs Outbound Mail

Inbound	Outbound
Internet → Gmail	Gmail → Internet
Depends on MX records	Uses SMTP
External sender	Internal sender

Remember:

MX affects **incoming** mail.

SMTP is used for **sending** mail.

10. Gmail Routing

Organizations often need custom routing.

Examples:

- Route mail to an external archive.
- Deliver to multiple mailboxes.
- Reject messages from specific domains.
- Add compliance recipients.
- Route based on departments.

Routing rules are configured in the Admin Console.

Enterprise Example

Company policy:

Every Finance email should also be archived.

Flow:

finance@contoso.com

↓

Google Routing Rule

↓

Finance Mailbox

+

Archive Mailbox

11. Common Email Problems

Cannot Receive External Email

Possible causes:

- Incorrect MX records
- DNS propagation
- Firewall or gateway issues
- Mail routing errors

SPF Failure

Possible causes:

- Sending server not listed in SPF
- Third-party email service missing
- Incorrect SPF syntax

DKIM Failure

Possible causes:

- Email modified in transit
 - Incorrect public key
 - DKIM not enabled
 - Signature mismatch
-

DMARC Failure

Possible causes:

- SPF failed
 - DKIM failed
 - DMARC policy set to reject
-

Mail Routed Incorrectly

Possible causes:

- Routing rule
 - Compliance rule
 - Content compliance policy
 - Gateway configuration
-

12. Troubleshooting Framework

When mail delivery fails,

follow a structured process.

1. Can the user send email?
2. Can the user receive internal email?
3. Can the user receive external email?
4. Check MX records.
5. Check SPF.
6. Check DKIM.
7. Check DMARC.
8. Review Gmail routing rules.
9. Review Email Log Search (or Email Log Search in the Admin Console) for message tracking.

Avoid changing DNS records without identifying the root cause.

Real Enterprise Scenario

Marketing uses:

- Google Workspace
- Mailchimp

Marketing campaigns are rejected.

Investigation:

- SPF authorizes only Google.
- Mailchimp is missing.

Resolution:

Update SPF to include Mailchimp's sending infrastructure.

Campaigns are delivered successfully.

This is exactly the example we covered together.

Best Practices

- ✓ Configure correct MX records.
 - ✓ Enable DKIM.
 - ✓ Publish an accurate SPF record.
 - ✓ Implement DMARC (start with `none`, then move to `quarantine` or `reject` after monitoring).
 - ✓ Review routing rules regularly.
 - ✓ Use message tracing when troubleshooting.
-

Common Mistakes

- ✗ Assuming SMTP controls incoming email.

- ✗ Forgetting to include third-party senders in SPF.
 - ✗ Ignoring DKIM failures.
 - ✗ Setting a strict DMARC policy before validating SPF and DKIM.
 - ✗ Changing multiple DNS records simultaneously without testing.
-

Interview Questions

1. What is the purpose of MX records?
 2. What does SMTP do?
 3. How does SPF prevent spoofing?
 4. How does DKIM verify email integrity?
 5. What is DMARC?
 6. Employees can send email but cannot receive external email. Where do you investigate first?
 7. Why would a Mailchimp campaign fail SPF validation?
 8. What happens when both SPF and DKIM fail and DMARC is set to reject?
-

Key Takeaways

- SMTP sends email.
 - DNS helps locate mail servers.
 - MX records control inbound mail delivery.
 - SPF authorizes sending servers.
 - DKIM verifies message integrity using digital signatures.
 - DMARC defines the action when authentication fails.
 - A structured troubleshooting process is essential for diagnosing mail flow issues.
-

Chapter Summary

In this chapter you learned:

- ✓ Email architecture.
- ✓ SMTP.
- ✓ DNS.
- ✓ MX records.
- ✓ SPF.
- ✓ DKIM.
- ✓ DMARC.
- ✓ Gmail routing.
- ✓ Inbound and outbound mail flow.
- ✓ Enterprise troubleshooting scenarios.

The next chapter will cover **Google Workspace Migration**, bringing together everything you've already practiced: migration strategies (cutover, phased, coexistence), migration tools such as **Google Workspace Migrate**, **Google Workspace Migration for Microsoft Exchange (GWMME)**, **Google Workspace Migration for Microsoft Outlook (GWMMO)**, **Data Migration Service (DMS)**, and **CloudM**, along with real-world migration planning and interview scenarios.